NHS England

Understanding and Reducing Missed GP Appointments: A Data-Driven Analysis of NHS Capacity and Resource Utilisation

Reshma Pandit

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NHS England

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Background

- The **NHS** seeks to understand and reduce the **financial and operational impact** of missed GP appointments across England.
- The NHS needs a data-informed approach to decide how to reduce missed appointments and optimise current operations by examining **appointment utilisation**, **staff capacity**, **Did Not Attend (DNA) rates** and **drivers of missed appointments**.
- Findings aim to support **targeted interventions** in planning, communication, and capacity management to improve efficiency and patient access

Business Questions

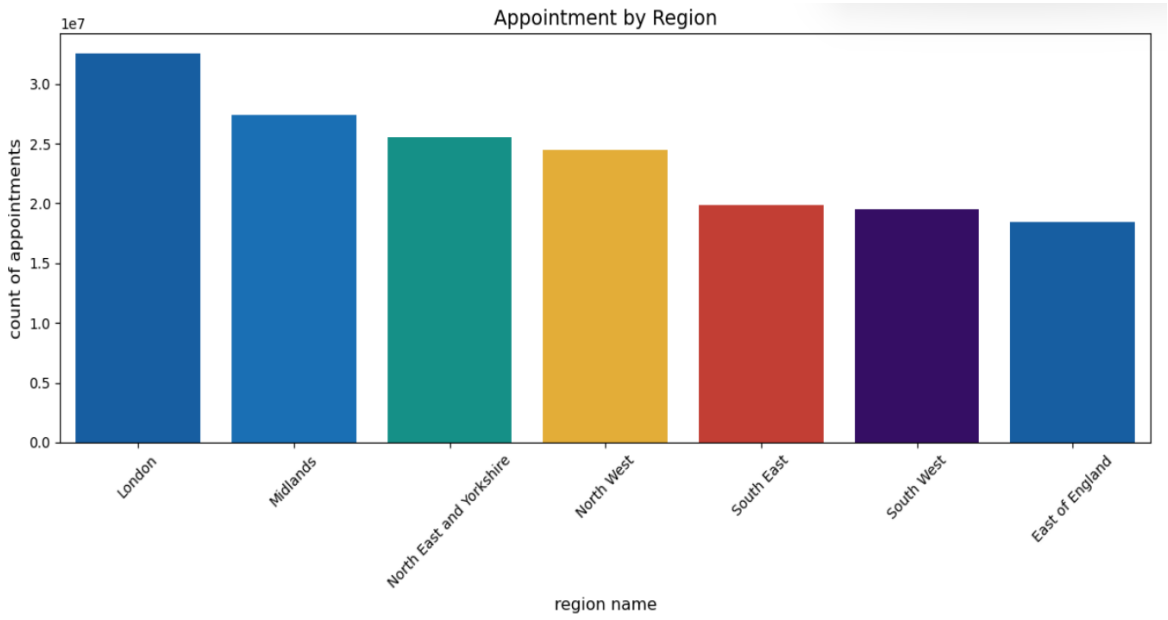
- Has there been adequate staff and capacity in the networks?
- What was the actual utilisation of resources?
- How could we reduce missed appointments to optimise current operations?

Analytical Approach

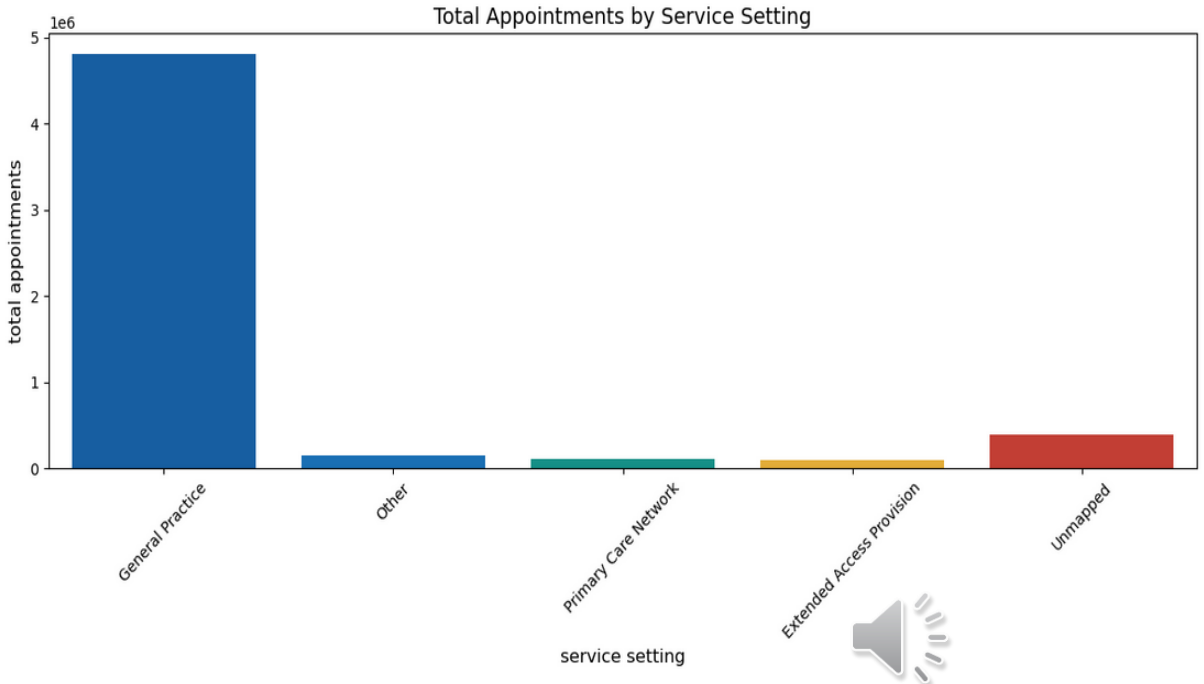
- Quantify **appointment activity across regional and service setting** to establish the baseline NHS utilisation in the networks.
- Measure variation in **appointment volumes over time** that could indicate **under or over-utilisation** of NHS capacity.
- Assess utilisation across **healthcare professional types** and **service settings** to identify care types that experience consistent pressure.
- Calculate and compare **Did Not Attend (DNA) rates** across **waiting periods** and **appointment modes** to highlight factors that contribute to missed appointments, and to support targeted interventions and communication strategies.
- Analyse **public sentiment** from Twitter data to uncover recurring themes about the NHS system that may influence attendance behaviour.

NHS England at a glance

- NHS England is spread across 7 regions, 42 ICBs and 106 sub-ICBs
- London is the busiest NHS region

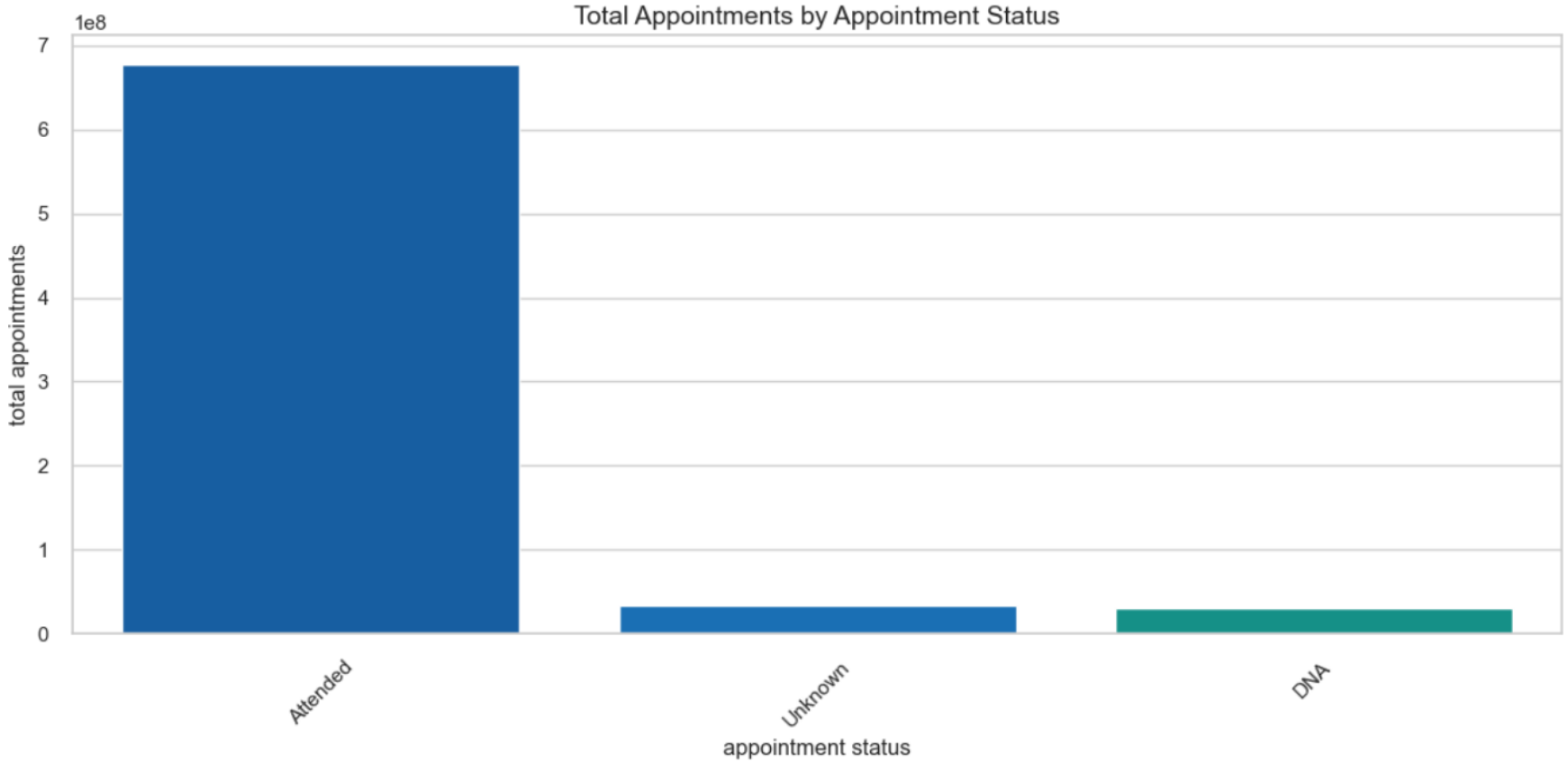


- NHS has a planned capacity to handle 1.2 million patients per day
- General Practice (GP) is the most utilised service setting as seen below.



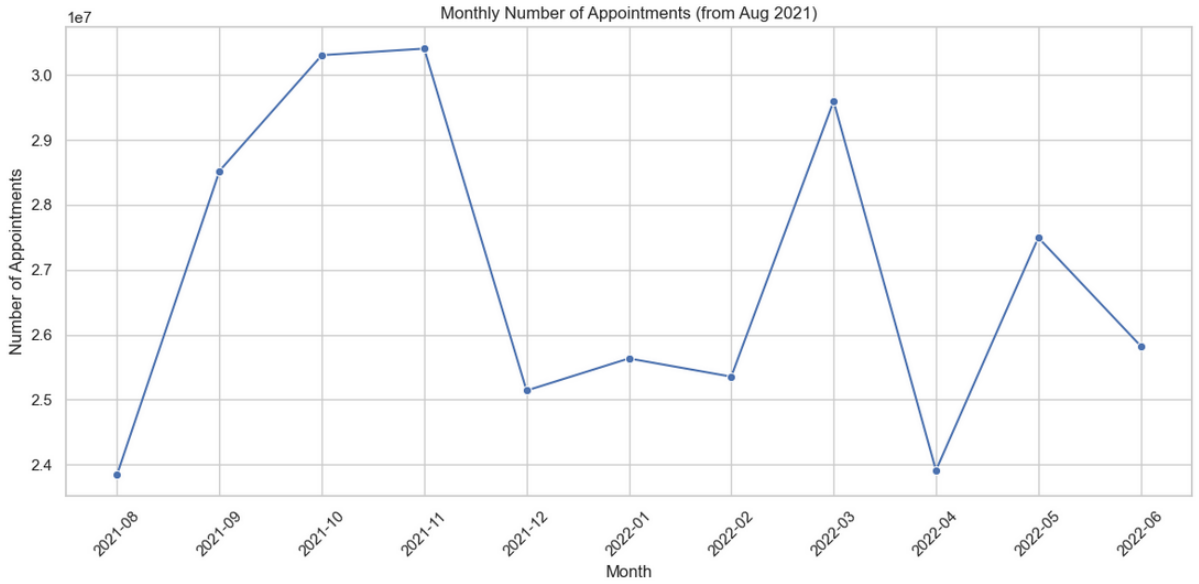
NHS at a glance

Reducing missed appointments could recover resources lost in approximately 4% (30911233 appointments for this dataset) of all scheduled appointments.

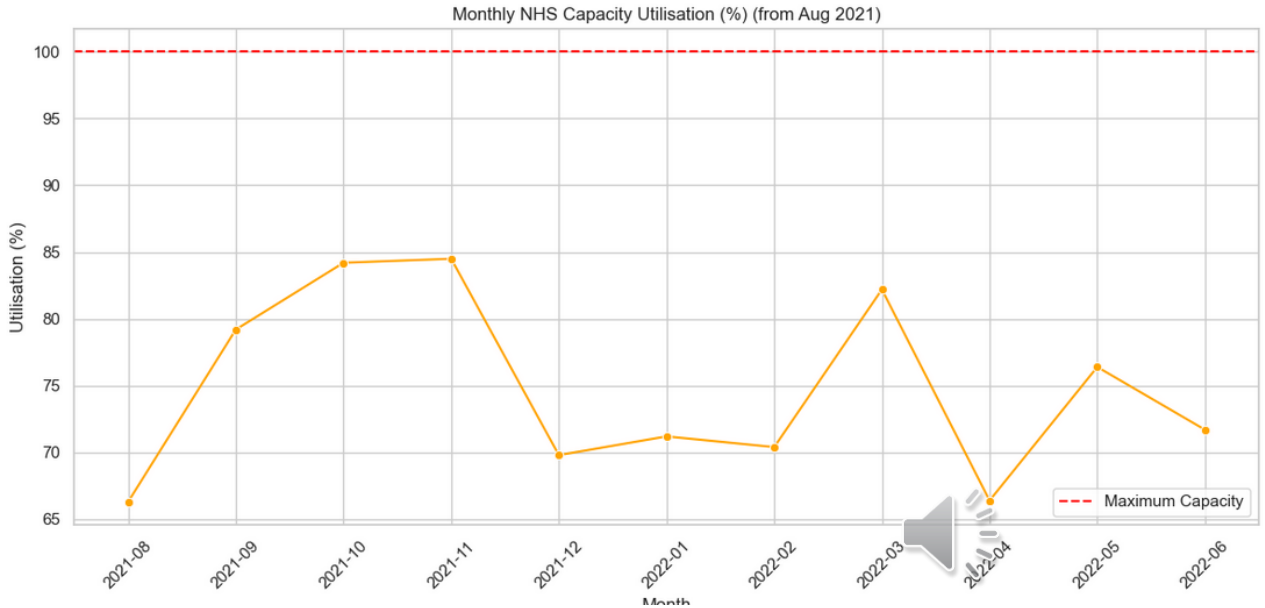


NHS utilisation in the networks

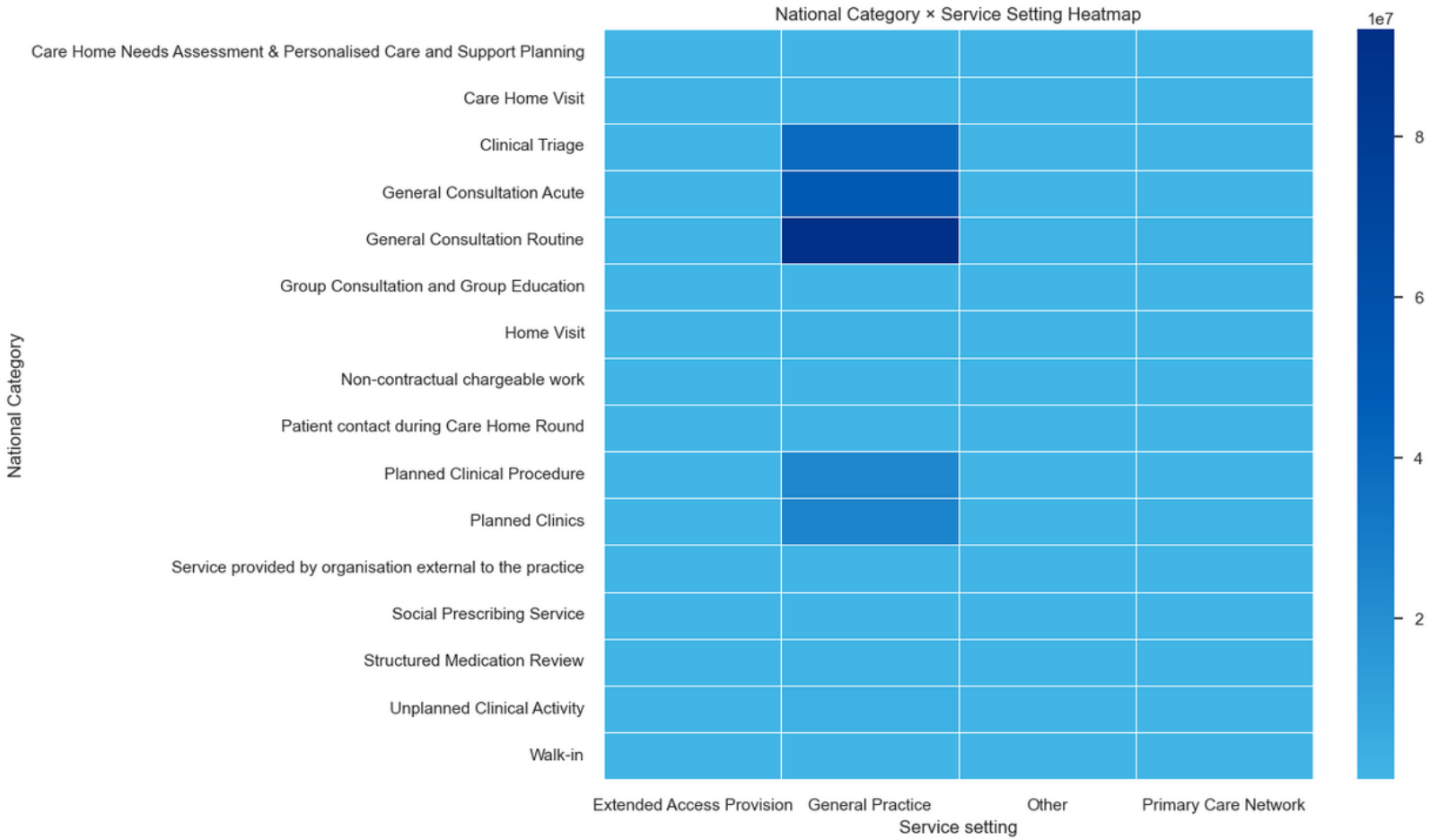
- Appointment volumes ranged from 23-31 million per month, peaking in October 2021 and March 2022
- This coincides with schools reopening after lifting of COVID-19 restrictions in 2021 and lifting of all COVID-19 restrictions in February 2022



- Despite high demand, utilisation reached up to ~85% of planned capacity
- Current system capacity is largely adequate but could be better balanced across time and regions.

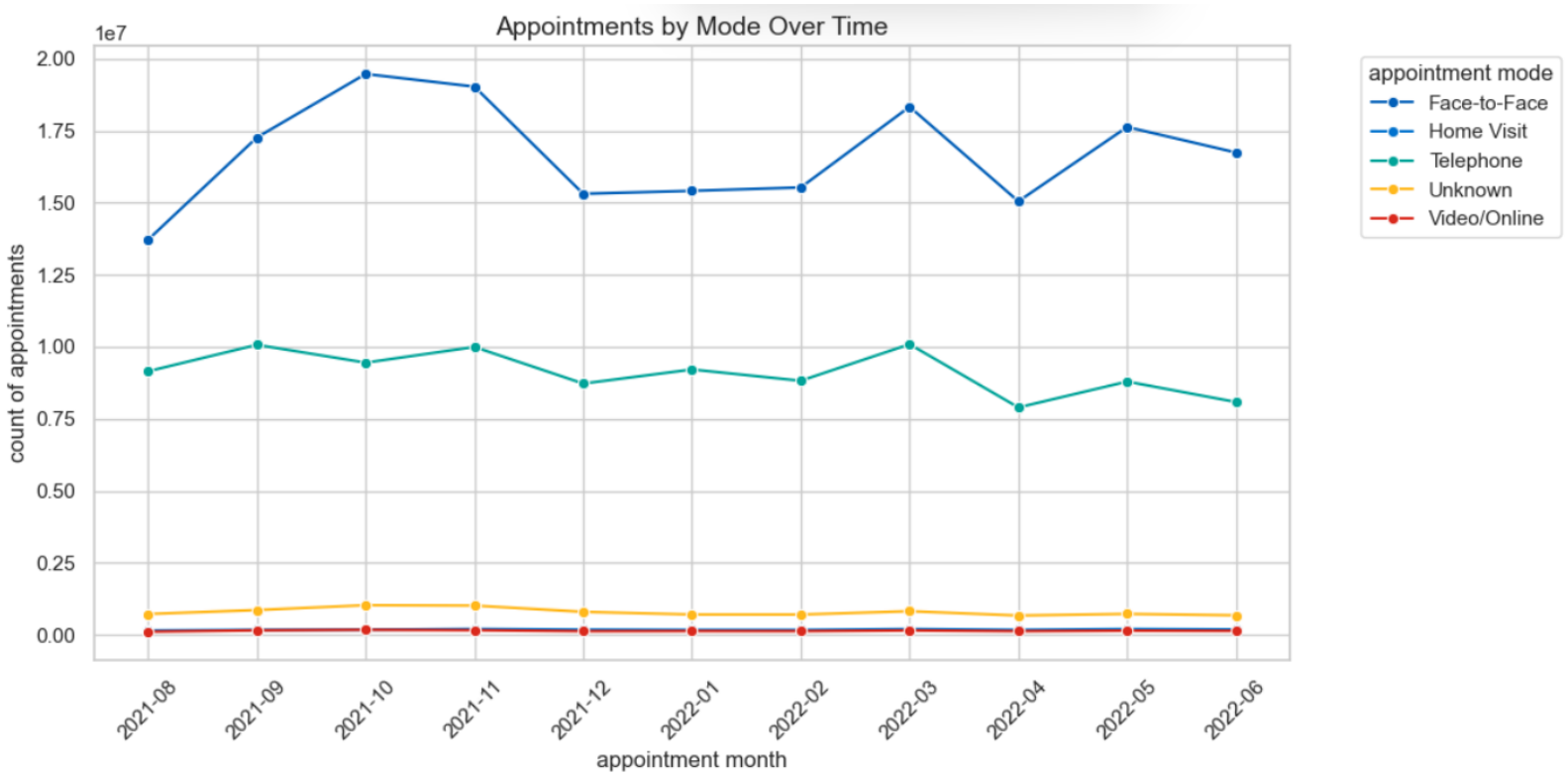


Utilisation across healthcare national categories and service settings



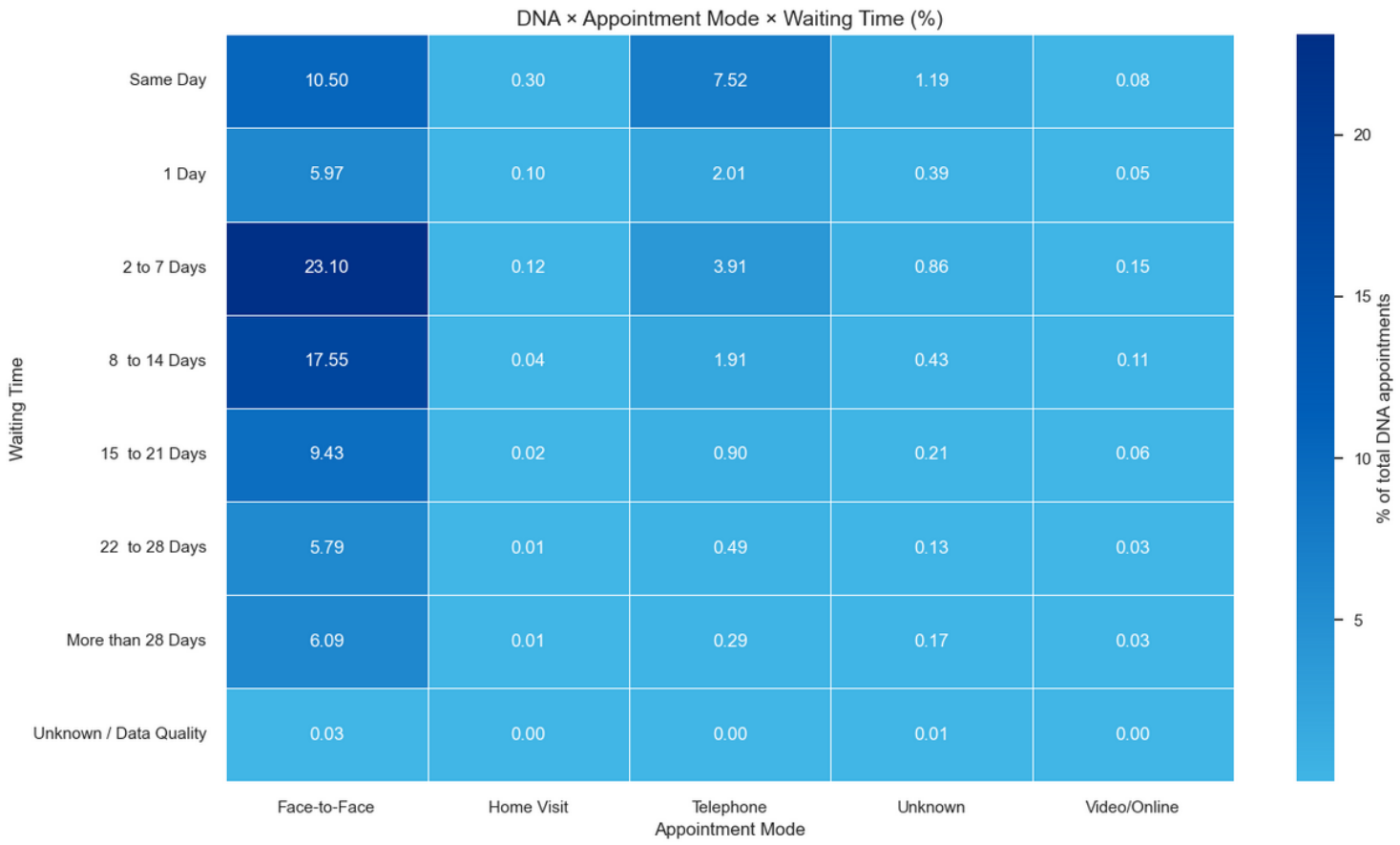
General practice shows the highest number of appointments across all service settings, with **general consultation (routine and acute), clinical triage and planned clinical procedures and clinics** driving the majority of appointments.

Utilisation across healthcare appointment modes



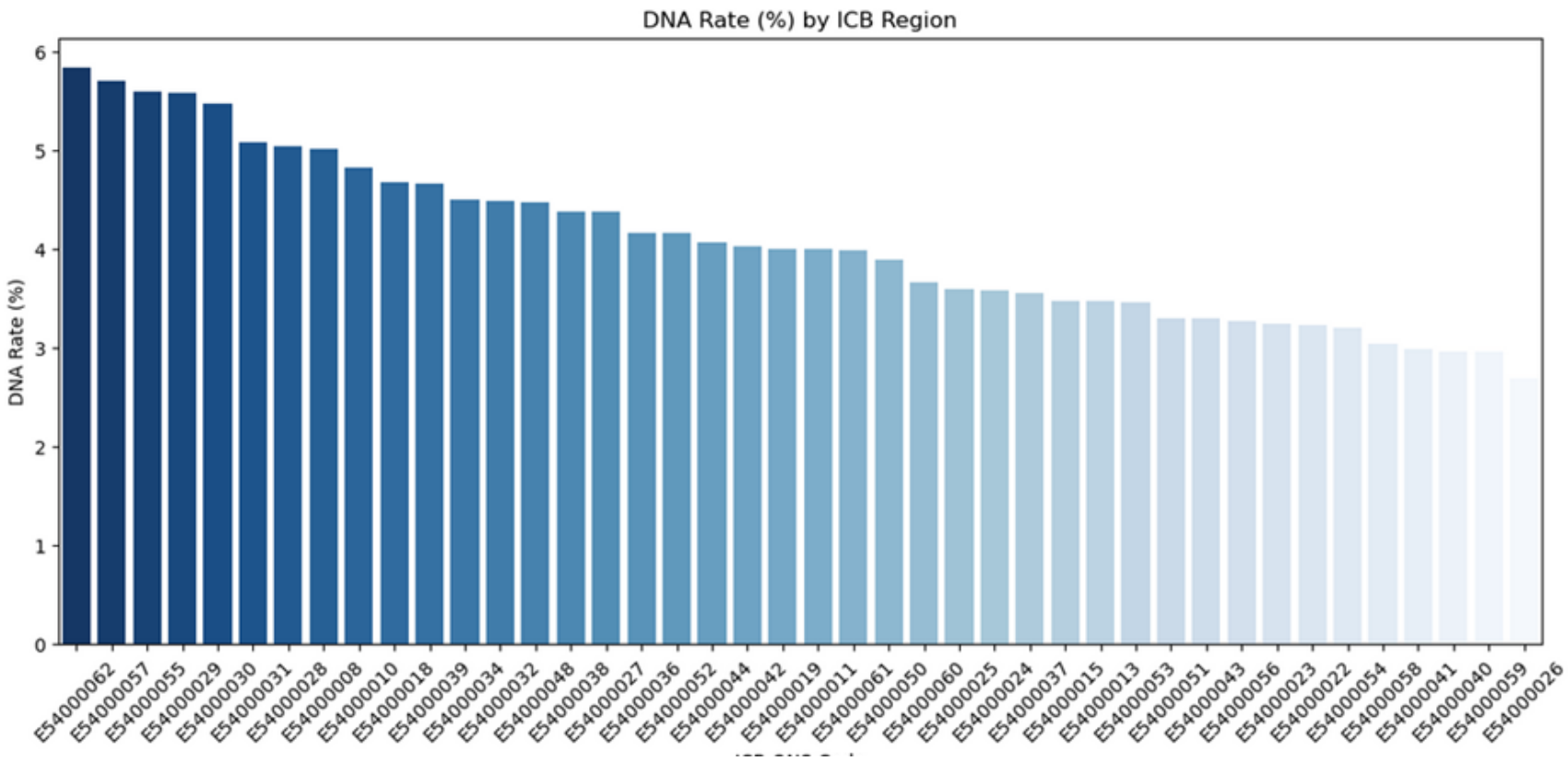
- **Face-to-face appointments** remain the dominant mode, absorbing most of the variability in demand (including October–November 2021 and March 2022).
- **Telephonic and video consultations remain steady and under-utilised.**
- This presents an opportunity to redistribute workload and **improve flexibility in service delivery,** especially in peak demand scenarios.

Did Not Attend (DNA) rates and factors that contribute to missed appointments



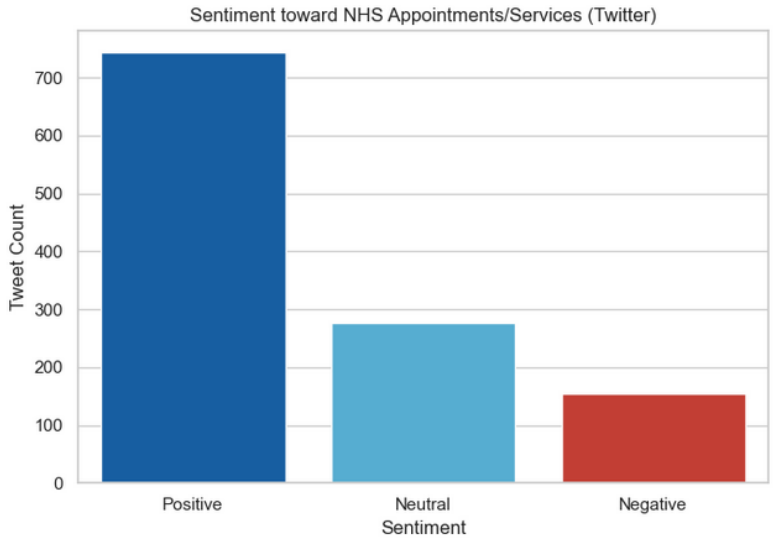
- **Face-to-face and same-day appointments show higher DNA rates**, possibly due to travel barriers or short notice.
- **Video/online and home visits show the lowest DNA rates, suggesting that flexible appointment** options and advance scheduling improve attendance.

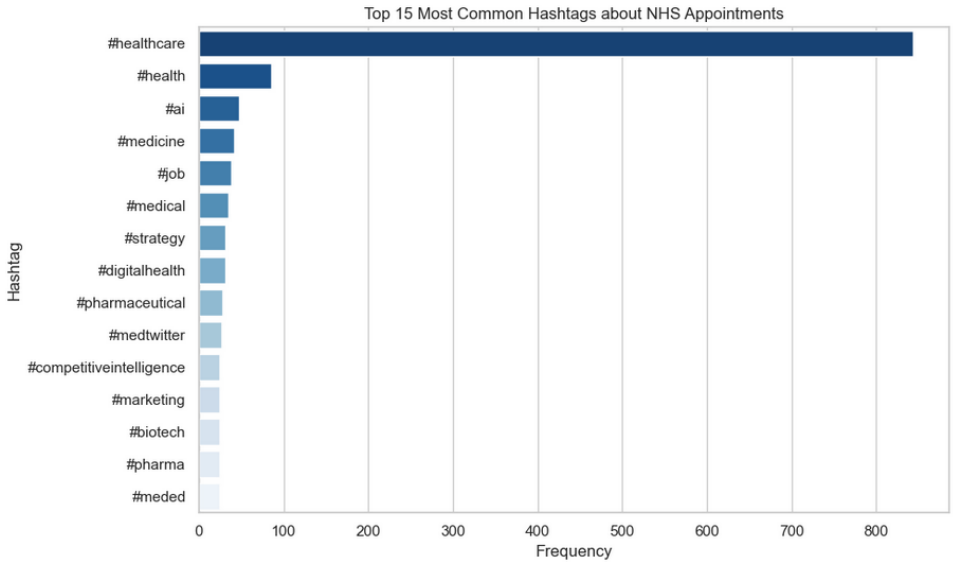
Did Not Attend (DNA) rates across regions



- DNA rates vary more than **twofold** across ICBs (from ~2.7% to ~5.8%).
- **Worst Performers**: Black Country , Greater Manchester, Birmingham & Solihull.
- **Best Performers**: Mid & South Essex , Northamptonshire , Bath & North East Somerset.
- These differences present opportunities for **behavioural or operational interventions** along with **learning opportunities** to reduce DNA appointments.

Assessing public sentiment from Twitter





- Twitter sentiment about the NHS was broadly **positive**
- There are minimal mentions of missed appointments or waiting-time complaints
- Public perception is not a major barrier to attendance

Recommendations

NHS England

1. Diversify Appointment Modes
(High Priority | Short-term: 0–6 months)
Why:
Variation in utilisation and high DNA rates suggest that appointment mode flexibility (in-person vs remote) directly affects attendance and efficiency.
Actions:
Expand digital and video consultation offerings, especially for follow-ups and non-urgent cases.
Pilot hybrid scheduling (patient choice of online/in-person mode).
Publicise remote options through GP practices and NHS app.
KPIs:
10–15% increase in remote consultation uptake within 6 months.
DNA rate reduction of at least 1% among remote appointment users.

2. Target High-DNA Regions for Intervention
(High Priority | Medium-term: 6–12 months)
Why:
ICBs such as **Black Country, Birmingham & Solihull, and Greater Manchester** consistently report above-average DNA rates, leading to inefficient use of capacity.

Actions:
Implement tailored reminder and rebooking systems (SMS/app).
Share best practices from low-DNA ICBs with high-DNA areas.

KPIs:
Reduce DNA rates in target ICBs by 2% within 12 months.
Recovery of ~1 million appointments annually through reduced DNAs.

Recommendations

3. Optimise Scheduling & Communication Systems
(Medium Priority | Short-term: 0–6 months)
Why:
Short-notice bookings and lack of flexible rebooking options contribute significantly to DNAs and uneven utilisation patterns.
Actions:
Extend lead time for routine appointments.
Automate reminders and enable self-service rescheduling via NHS app.
Introduce same-day cancellation reallocation system.
KPIs:
Increase appointment confirmation rate by 5% within 6 months.
Reduce short-notice appointment cancellations by 10%.

4. Strengthen Data Infrastructure and Quality
(High Priority | Long-term: 12–24 months)
Why:
Inconsistent appointment data entry and missing operational fields limit accurate performance monitoring and policy evaluation.
Actions:
Standardise appointment data entry formats across ICBs.
Automate validation checks and missing data alerts.
Integrate appointment data with operational and workforce dashboards.
KPIs:
100% of ICBs using standardised data templates within 18 months.
90% completeness rate for key appointment data fields.

Recommendations

5. Strengthen Data Infrastructure:
(Medium Priority | Ongoing)

- **Why:**
  Ongoing visibility into utilisation, DNA trends, and appointment modes enables proactive policy and operational interventions.

- **Actions:**
- Develop real-time utilisation dashboards by ICB, month, and mode.
- Track performance against capacity, DNA rates, and service type.
- Share quarterly insight reports with NHS regional leads.

- **KPIs:**
- Quarterly dashboard adoption by all ICBs.
- Performance reports generated within 2 weeks of data refresh.

Thank you!